Lecture 1: AI/ML Foundations and Closed-Loop Control for Particle Accelerators

Physics Foundations, Neural Surrogates, Reinforcement Learning, and Bayesian Optimization

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KOREA UNIVERSITY

WORKSHOP CURRICULUM | # Integrated Morning Lectures & Afternoon Hackathon

Bridging theoretical foundations directly with closed-loop simulation and live W&B tracking

Lecture 1 (09:10 – 10:30)

Instructor: Chong Shik Park, Ph.D.

- Physics Foundations & Hill's Equation
- Orbit Response Matrix (ORM) Coupling
- Non-Linear Cubic Core Saturation
- Neural Surrogate Digital Twins (`MLP`)
- SVD Pseudo-Inverse vs Bounded Inversion
- Physics-Informed RL & Dynamic Damping
- Gaussian Process Bayesian Optimization

Target: Sub-micron closed-loop control under non-linear actuator constraints.

Lecture 2 (10:40 – 12:00)

Instructor: Thorsten Hellert, Ph.D.

- Agentic AI Frameworks in Accelerators
- Large Language Model (LLM) Agents
- *Osprey* Autonomous Control System
- Automated Scientific Workflows
- Python API Tool-Calling Architecture
- Operational Guardrails & Safety Bounds
- Human-in-the-Loop Supervision

Target: Autonomous tuning agents and scientific workflow orchestration.

Mini-Hackathon (13:30 – 17:00)

Hands-On Orbit Alignment Challenge

- 15-meter Transport Lattice (12 BPMs, 6 Steerers)
- 3 Parallel Tracks (MLP, RL, BO)
- Standardized 30-Seed Evaluation Suite
- Sub-30 μm Convergence Target
- Real-Time Weights & Biases (W&B) Logging
- Live Global Leaderboard Tracking
- Final Awards & Recognition

Target: $> 1,000$ points on standardized evaluation suite.

Five Foundational Pillars of AI in Modern Accelerators

How artificial intelligence transforms design, commissioning, and autonomous operations

1. Neural Surrogate Digital Twins

- Replaces slow tracking codes (`ELEGANT`, `MAD-X`) with sub-millisecond deep neural surrogates.
- Enables real-time model-predictive control, online orbit feedback, and virtual parametric sensitivity analysis.

2. Multi-Objective Bayesian Optimization

- Balances trade-offs: beam transmission vs photon brightness vs transverse emittance.
- Gaussian Processes with acquisition functions find Pareto-optimal set in dozens of shots instead of thousands.

3. Virtual Diagnostics & Phase Space Reconstruction

- Reconstructs destructive beam metrics (6D phase space, bunch slice emittance) non-invasively.
- Neural networks predict shot-by-shot profiles from upstream RF cavities and BPM telemetry.

4. Anomaly Detection & Machine Protection

- Unsupervised models monitor sensor drift to prevent multi-megawatt beam downtime.
- Deep sequence models predict RF breakdown and cavity quenches milliseconds before catastrophic trips.

5. Agentic AI & Autonomous Operations

- Integrates LLM reasoning engines with accelerator APIs and diagnostic archives (e.g. *Osprey*).
- Executes high-level operational commands safely within strict hardware interlock boundaries.

Transverse Beam Dynamics & Hill's Equation

Theoretical governing equation shaping betatron oscillations in linear magnetic lattices

Hill's Equation Formulation

Transverse particle motion along path s in an uncoupled linear channel:

$$\frac{d^2x(s)}{ds^2} + K(s)x(s) = 0$$

- $K(s) = \frac{e}{p}\frac{\partial B_y(s)}{\partial x}$ represents quadrupole focusing ($K > 0$ for QF, $K < 0$ for QD).
- General solution is a pseudo-harmonic **betatron oscillation**:

$$x(s) = \sqrt{\epsilon\beta(s)}\cos(\psi(s) + \phi_0)$$

- $\beta(s)$: Courant-Snyder beta function (transverse beam envelope)
- $\psi(s) = \int_0^s \frac{d\bar{s}}{\beta(\bar{s})}$: Betatron phase advance
- $\nu = \frac{1}{2\pi}\oint \frac{ds}{\beta(s)}$: Betatron tune (total phase advance in units of 2π)

The Operational Challenge: Hadron Linacs & Rings

Sub-Micron Stability Tolerances:

- Modern light sources require orbit stabilization to $< 10\%$ of beam size (sub-micron RMS tolerances).
- In multi-megawatt hadron linacs (e.g. KOMAC, SNS, ESS):
  - Offsets in quadrupoles/cavities induce strong space charge and halo.
  - Losses exceeding 1 W/m (10^{-5} of total current) cause severe structural radioactivation.
  - Hands-on maintenance is prohibited if orbit deviations remain uncorrected.

Physical Disturbance Mechanisms:

- Ground motion and mechanical vibration (pumps, cryocoolers)
- Thermal girder expansion and diurnal ambient temperature cycles
- Power supply current ripple and magnet core hysteresis

Derivation of the Orbit Response Matrix (ORM)

Mathematical link connecting steerer deflections to downstream diagnostic readings

Theoretical ORM Derivation

Localized angular kick θ_j applied by steering dipole j:

$$R_{ij} = \frac{\Delta x(s_i)}{\theta_j} = \frac{\sqrt{\beta_i \beta_j}}{2\sin(\pi\nu)} \cos(|\psi_i - \psi_j| - \pi\nu)$$

Linear Superposition Formulation:

$$\mathbf{Y}_{\text{linear}} = \mathbf{I} \cdot \mathbf{R}$$

- $\mathbf{I} \in \mathbb{R}^6$: Vector of 6 corrector currents in Amperes
- $\mathbf{R} \in \mathbb{R}^{6\times 12}$: Linear Orbit Response Matrix
- $\mathbf{Y} \in \mathbb{R}^{12}$: Transverse displacements at 12 BPM stations

Coupling Dynamics:
Deflecting a single steerer perturbs the entire downstream trajectory through betatron wave propagation.

Why Pure Linear Inversion Breaks Down

Classical Singular Value Decomposition (SVD):

$$\Delta\mathbf{I} = -\alpha \cdot (\mathbf{R}^T)^\dagger \cdot \mathbf{Y}$$

Key Vulnerabilities in Real-World Operations:

1. **Magnetic Core Saturation:**
High coil currents drive iron yokes into non-linear saturation. The true response departs significantly from $\mathbf{R}$.

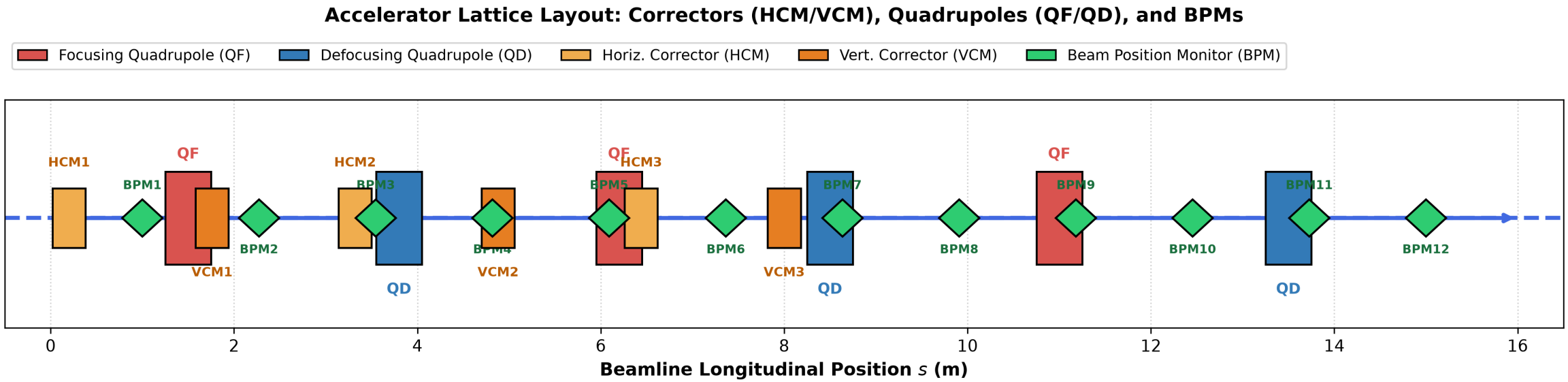
2. **Limit Cycles & Overshoot:**
High static feedback gains ($\alpha > 0.5$) cause severe overshoot and persistent oscillatory limit cycles in saturated regimes.

3. **Diagnostic Sensor Noise:**
Sensor jitter ($\sigma = 10$ μm) is inverted as false physical errors, commanding unnecessary actuator kicks.

HACKATHON BENCHMARK |

The 15-Meter Transport Lattice Architecture

Layout of quadrupole doublets, dipole steering magnets, and beam position monitors



Diagnostics & Actuators

- **12 Diagnostic BPMs:** Evenly spaced from $s = 1.0$ m to $s = 15.0$ m.
- **6 Dipole Steering Magnets:** 3 Horizontal (HCM1-3) & 3 Vertical (VCM1-3).
- **Quadrupole Doublets:** Alternating QF & QD magnets shaping transverse beta.

Hardware Constraints

- **Cumulative Current Bounds:** $\mathbf{I} \in [-5.0\text{ A}, +5.0\text{ A}]$ (thermal limit).
- **Single-Step Delta Bounds:** $\Delta\mathbf{I} \in [-0.8\text{ A}, +0.8\text{ A}]$ (rate of change limit).
- **Sensor Noise Floor:** Gaussian jitter $\boldsymbol{\epsilon} \sim \mathcal{N}(0, 0.010^2\text{ mm})$ (10 μm RMS).

Non-Linear Magnetic Saturation & Misalignments

Modeling realistic iron-core saturation and random quadrupole alignment errors

Mathematical Response Formulation

Physical Beam Position Monitor Reading $\mathbf{Y}$:

$$\mathbf{Y} = \mathbf{I} \cdot \mathbf{R} - 0.015 \cdot (\mathbf{I}^3 \cdot \mathbf{R}) + \mathbf{Y}_0 + \boldsymbol{\epsilon}$$

Components Breakdown:

1. $\mathbf{I} \cdot \mathbf{R}$: Pure linear betatron response
2. $-0.015 \cdot (\mathbf{I}^3 \cdot \mathbf{R})$: Non-linear iron core soft saturation
   - At $I = 2.0$ A: 6% drop in effective steerer authority
   - At $I = 4.0$ A: 24% authority loss (linear model severely overshoots)
3. $\mathbf{Y}_0$: Initial orbit distortion from misaligned quadrupoles
4. $\boldsymbol{\epsilon} \sim \mathcal{N}(0, 0.010^2$ mm): Diagnostic Gaussian jitter (10 μm)

Initial Distortion & Noise Profiles

Initial Distortion Generation ($\mathbf{Y}_0$):

Quadrupole misalignments induce harmonic betatron waves:

$$Y_{0,k} = A_1 \sin\left(\frac{2\pi s_k}{\lambda_1} + \phi_1\right) + A_2 \cos\left(\frac{2\pi s_k}{\lambda_2} + \phi_2\right)$$

- Produces realistic initial RMS orbit errors between 0.40 mm and 1.10 mm across evaluation seeds.

The Benchmark Convergence Target:

Target RMS: < 0.030 mm (30 μm)

To achieve success, competitor policies must suppress initial orbit distortion by **93% to 97%** within a strict budget of 20 control steps.

Deep Neural Networks & Supervised Learning Foundations

Architectural principles of artificial neurons, non-linear activations, and loss optimization

Anatomy of an Artificial Neural Network

Artificial Neuron Transformation:

$$z = \sum_j w_j x_j + b = \mathbf{w}^T \mathbf{x} + b, \quad a = \sigma(z)$$

Multi-Layer Perceptron (MLP):

Chaining L affine transformations with element-wise non-linear activations:

$$\mathbf{h}^{(l)} = \sigma\left(\mathbf{W}^{(l)} \mathbf{h}^{(l-1)} + \mathbf{b}^{(l)}\right)$$

Activation Function: Rectified Linear Unit (ReLU)

$$\sigma(z) = \max(0, z)$$

- Constant gradient $= 1$ for $z > 0$ (prevents vanishing gradients)
- Computationally ultra-efficient; introduces piecewise linearity

Loss Formulation & Adam Optimizer

Objective Function (MSE + L2 Weight Decay):

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{N} \sum_{i=1}^{N} \|\mathbf{y}_i - f_{\boldsymbol{\theta}}(\mathbf{I}_i)\|^2 + \lambda \|\boldsymbol{\theta}\|^2$$

Adam (Adaptive Moment Estimation):

- Tracks first moment $\mathbf{m}_t$ and uncentered second moment $\mathbf{v}_t$:

$$\mathbf{m}_t = \beta_1 \mathbf{m}_{t-1} + (1 - \beta_1)\mathbf{g}_t, \quad \mathbf{v}_t = \beta_2 \mathbf{v}_{t-1} + (1 - \beta_2)\mathbf{g}_t^2$$

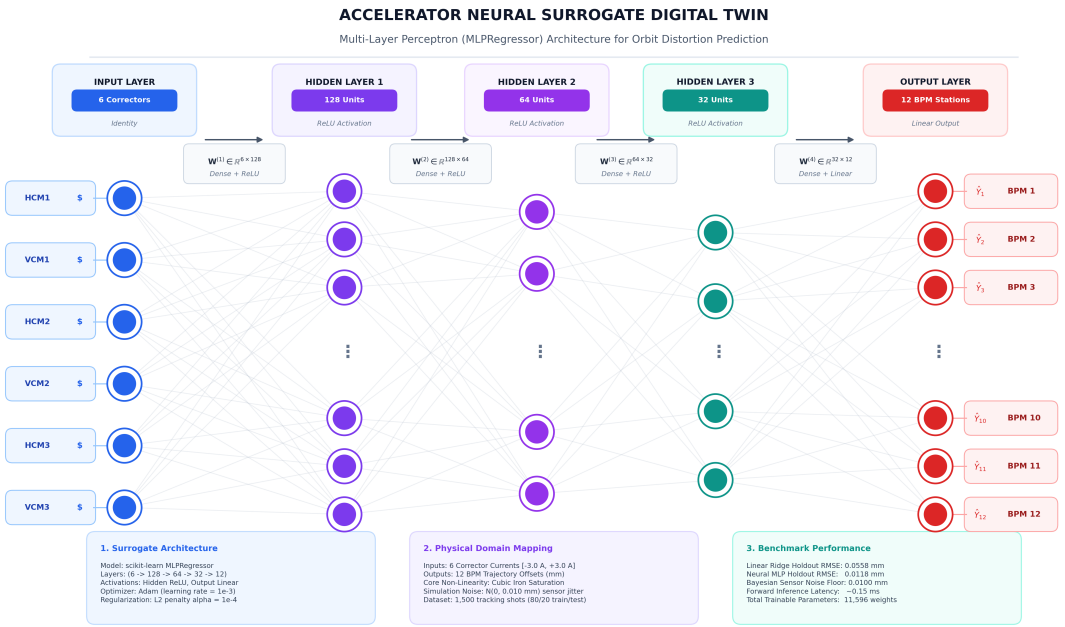
- Parameter update with bias correction:

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \frac{\eta}{\sqrt{\hat{\mathbf{v}}_t} + \epsilon} \hat{\mathbf{m}}_t$$

- Provides robust convergence across varying gradient scales without requiring manual per-parameter learning rate tuning.

Designing the Neural Surrogate Digital Twin

Capturing non-linear cubic saturation with MLPRegressor for sub-millisecond execution



Surrogate Architecture & Training

Dataset Generation:

- 2,000 synthetic tracking runs sampling corrector currents uniformly across $\mathbf{I} \in [-3.0, 3.0]^6$ A.
- Accurately captures both linear optics and cubic core saturation.

MLPRegressor Hyperparameters:

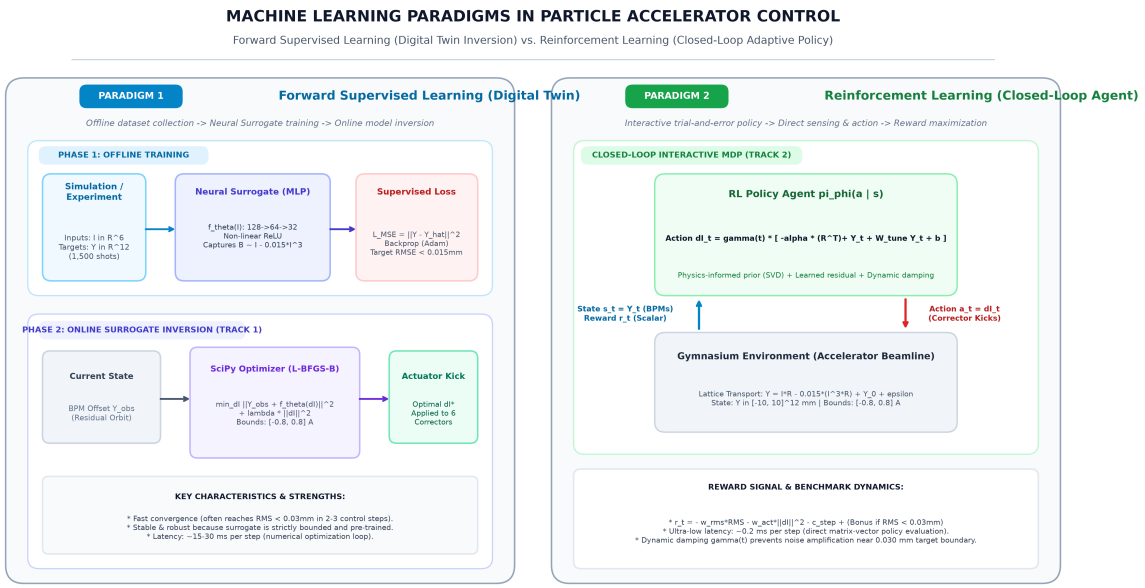
- **Input:** 6 Corrector currents [HCM1-3, VCM1-3]
- **Hidden Architecture:** $(128, 64, 32)$ neurons
- **Activation:** ReLU; **Solver:** Adam
- **Regularization:** $\alpha_{\text{L2}} = 10^{-4}$, `max_iter=500`
- **Output:** 12 predicted BPM readings [BPM1-12]

Performance Verification:

- Holdout Test MSE: $< 2.5 \times 10^{-4}$ mm^2
- Forward inference latency: < 0.2 ms per call

Forward Prediction vs Closed-Loop Orbit Control

Transitioning from predictive surrogate twins to continuous closed-loop feedback policies



Closed-Loop Control Dynamics

The Closed-Loop Feedback Cycle:

1. **Diagnostics:** 12 Beam Position Monitors measure current transverse orbit error $\mathbf{Y}_t \in \mathbb{R}^{12}$.
2. **Policy Decision:** Orbit controller computes corrective kick vector $\Delta\mathbf{I}_t \in [-0.8, 0.8]^6$ A.
3. **Actuator Execution:** Power supplies update cumulative magnet currents:
$$\mathbf{I}_{t+1} = \mathbf{clip}(\mathbf{I}_t + \Delta\mathbf{I}_t, -5.0, 5.0)\ \mathbf{A}$$
4. **Lattice Response:** Corrected magnetic fields deflect beam; new transverse orbit $\mathbf{Y}_{t+1}$ is established.
5. **Termination:** Succeeds if RMS < 0.030 mm or truncates if step count reaches 20.

Baseline 0 (Linear SVD) vs Baseline 1 (Neural Inversion)

Comparing classical pseudo-inverse control with model-based bounded optimization

Baseline 0: Classical Linear SVD

Mathematical Formulation:

$$\Delta \mathbf{I} = -\alpha \cdot (\mathbf{R}^T)^\dagger \cdot \mathbf{Y}$$

Operational Characteristics:

- Static gain $\alpha \approx 0.45$, hard clipped to $[-0.8, 0.8]$ A
- Execution latency: < 0.1 ms (instantaneous)

Strengths & Limitations:

✓ Guaranteed convergence in purely linear regimes
× Completely blind to cubic core saturation
× Causes overshooting and limit cycles at high currents
× Sluggish convergence (Mean steps: 5.6)

Official Score: 1,013.1 points (Rank #3)

Baseline 1: Supervised Neural Inversion

Model-Based Optimization:

$$\min_{\Delta \mathbf{I}} \|\mathbf{Y}_{\mathbf{obs}} + f_{\boldsymbol{\theta}}(\Delta \mathbf{I})\|^2 + \lambda \|\Delta \mathbf{I}\|^2$$
$$\text{subject to } \Delta \mathbf{I} \in [-5.0, 5.0]^6 \text{ A}$$

Operational Characteristics:

- Solver: SciPy `L-BFGS-B` on trained MLP digital twin
- Execution latency: $\sim 15 - 25$ ms per step

Why It Outperforms Classical SVD:

✓ Accurately anticipates non-linear core saturation
✓ Computes optimal kicks in non-linear regimes
✓ Reaches convergence in just 2.3 steps average

Official Score: 1,016.8 points (Rank #2)

Closed-Loop RL with Physics-Informed Priors

Continuous policy networks combining analytical optics priors with dynamic step damping

Physics-Informed Residual Architecture

Residual Policy Formulation:

$$\Delta\mathbf{I}(t) = \gamma(t) \cdot \left[-\alpha(\mathbf{R}^T)^\dagger \mathbf{Y} + \mathbf{W}_{\mathbf{learned}}\mathbf{Y} + \mathbf{b}\right]$$

Two Core Synergistic Components:

1. **Analytical Physics Prior ($-\alpha(\mathbf{R}^T)^\dagger\mathbf{Y}$):**
- Guarantees immediate coarse stabilization from step 1.
- Prevents unguided random exploratory actions.

2. **Learned Residual Offset ($\mathbf{W}_{\mathbf{learned}}\mathbf{Y} + \mathbf{b}$):**
- Compensates for non-linear saturation deficit.
- Learns asymmetric corrections across HCM and VCM plane.

Dynamic Harmonic Step Damping

The Sensor Noise Amplification Problem:
Near the 0.030 mm boundary, sensor noise ($\epsilon \sim 10\ \mu$m) creates spurious apparent gradients, triggering limit-cycle oscillations.

Harmonic Damping Schedule:

$$\gamma(t) = \frac{1.0}{1.0 + \kappa \cdot t}, \qquad \kappa \approx 0.03$$

- **Step 1 ($t = 0, \gamma = 1.0$):** Full kick authority rapidly eliminates millimeter-scale initial distortions.
- **Step 5+ ($t \geq 4, \gamma < 0.88$):** Attenuated feedback prevents sensor noise amplification and locks RMS below 0.030 mm.

TRACK 3 CHALLENGE |

Closed-Loop Bayesian Optimization with Gaussian Processes

Sample-efficient probabilistic modeling with Lower Confidence Bound (LCB) acquisition

Probabilistic Gaussian Process Modeling

GP Mapping $\Delta\mathbf{I} \mapsto \mathbf{Y}$ Orbit Deviation:

$$\hat{\mathbf{Y}}(\Delta\mathbf{I}) \sim \mathcal{GP}\left(\mathbf{m}(\Delta\mathbf{I}), \mathbf{K}(\Delta\mathbf{I}, \Delta\mathbf{I}')\right)$$

Composite Kernel Architecture:

$$k(\mathbf{x}, \mathbf{x}') = k_{\textbf{Matérn-5/2}}(\mathbf{x}, \mathbf{x}') + \sigma^2_{\textbf{noise}}\delta(\mathbf{x}, \mathbf{x}')$$

- **Matérn-5/2 Kernel:** Captures smooth, non-linear iron core saturation dynamics without unrealistic infinite differentiability.
- **White Noise Kernel:** Explicitly models the 10 μm BPM diagnostic jitter floor.

Lower Confidence Bound (LCB) Acquisition

LCB Acquisition Function:

$$\alpha_{\textbf{LCB}}(\Delta\mathbf{I}) = \textbf{RMS}\left(\mathbf{Y}_{\textbf{obs}} + \boldsymbol{\mu}_{\textbf{GP}}\right) - \kappa \cdot \overline{\boldsymbol{\sigma}}_{\textbf{GP}} + \lambda\|\Delta\mathbf{I}\|^2$$

Exploration vs Exploitation Trade-off:

- $\boldsymbol{\mu}_{\text{GP}}(\Delta\mathbf{I})$: Posterior mean predicted orbit
- $\overline{\boldsymbol{\sigma}}_{\text{GP}}(\Delta\mathbf{I})$: Posterior epistemic uncertainty
- $\kappa \geq 0$: Exploration bonus coefficient

Benchmark Outcome:

When calibrated, Track 3 achieves optimal convergence in just **1.5 steps average**, with the highest current economy.

Gymnasium Contract & Official Scoring Formula

Standardized Farama Gymnasium environment interface and multi-objective performance metric

Gymnasium Contract (`AcceleratorOrbitGymEnv`)

Environment Interface:

```
obs, info = env.reset(seed=s)
obs, reward, terminated, truncated, info = env.step(action)
```

- **obs:** np.ndarray shape $(12,)$ in $[-10, 10]$ mm
- **action:** np.ndarray shape $(6,)$ clipped to $[-0.8, 0.8]$ A
- **Step reward:**

$$r_t = -\text{RMS}_{t+1} - 0.01\|\Delta\mathbf{I}\|_2 + (50.0 \text{ if terminated})$$

- **Termination:** RMS < 0.030 mm (success!)
- **Truncation:** step ≥ 20 (failed to converge)

Official Hackathon Scoring Formula

Score Formula (Across 30 Standardized Seeds):

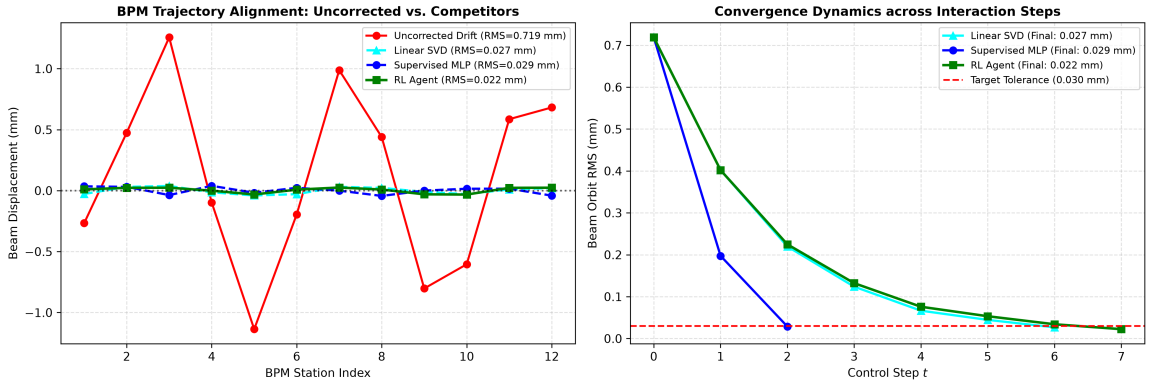
$$\textbf{Score} = (10 \times \textbf{Success\%}) - (2.0 \times \overline{\textbf{Steps}}) + \left(\frac{0.03}{\max(\overline{\textbf{RMS}}, 0.005)} \times 20\right) - (2.0 \times \overline{\|\mathbf{I}\|})$$

Four Scoring Components:

1. **Success Rate** $(10 \times \%)$**:** Up to $+1,000$ pts (100% convergence)
2. **Speed Penalty** $(-2 \times \overline{\textbf{Steps}})$**:** -2 pts per step taken
3. **Sub-Micron Precision Bonus:** Up to $+120$ pts for ultra-low residual RMS
4. **Current Economy Penalty:** -2 pts per Ampere of cumulative current norm

30-Seed Benchmark Leaderboard & Trajectories

Comparison of Classical SVD, Supervised MLP, Closed-Loop RL, and Bayesian Optimization



Official Benchmark Leaderboard

[Rank 1] Bayesian Opt (Track 3)

- Success: **100.0%** | Steps: **1.5 steps**
- Final RMS: **0.0149 mm** | Current: **0.93 A**
- Challenge Score: **1,035.4 pts**

[Rank 2] Supervised MLP (Baseline 1)

- Success: 100.0% | Steps: 2.3 steps
- Final RMS: 0.0257 mm | Current: 1.21 A
- Challenge Score: 1,016.8 pts

[Rank 3] Classical SVD (Baseline 0)

- Success: 100.0% | Steps: 5.6 steps
- Final RMS: 0.0230 mm | Current: 1.16 A
- Challenge Score: 1,013.1 pts

[Rank 4] Closed-Loop RL (Track 2)

- Success: 100.0% | Steps: 6.1 steps
- Final RMS: 0.0239 mm | Current: 1.18 A
- Challenge Score: 1,011.1 pts

Key Takeaways, Practical Advice & Next Steps

Strategic advice for afternoon hackathon competitors and transition to Lecture 2

Key Takeaways from Lecture 1

1. **Physics Guides the AI:**
Never train from scratch when the analytical ORM is known. Using pseudo-inverse priors guarantees early stability.

2. **Surrogates Capture Non-Linearities:**
When iron yokes saturate, compact MLPs accurately represent cubic falloff and enable 2-step bounded inversions.

3. **Probabilistic Uncertainty Wins:**
Gaussian Process BO balances exploration against exploitation, leading the 30-seed leaderboard in 1.5 steps.

4. **Adaptive Damping Locks Stability:**
Attenuating feedback near the 10 μm noise floor prevents sensor-induced limit cycles.

Practical Hackathon Advice & Next Session

Practical Competitor Advice:

- **If your policy oscillates:** Check feedback gain α. Introduce harmonic damping $\gamma(t) = 1/(1 + 0.03t)$.
- **If surrogate error is high:** Ensure training data covers $[-3.0, 3.0]$ A with cubic non-linearities.
- **Log Telemetry to W&B:** Compare your run curves against the official benchmark on the live leaderboard.

Next Session (10:40 – 12:00):

Lecture 2: Agentic AI for Accelerator Research (Osprey Applications)

Speaker: Thorsten Hellert, Ph.D. (LBNL)